HRRP

Design Catalogue for Reconstruction of Earthquake Resistant House
Volume II

Technical Coordination Meeting
Dhading Besi, Dhading
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Housing Recovery & Reconstruction Platform (HRRP)

Introduction

- DUDBC has prepared the document.
- It covers seventeen model design following twelve technologies.
- The designs provided in this catalogue are based on calculations, model test and analytical tests as these technologies are not covered by Nepal National Building Code, 2060.
- It covers mainly Alternative Construction Materials and Technologies.

Alternative Construction Materials and Technologies.

- Should be Locally available
- Production and correction on construction method from local People.
- Strong, Cheap & durable.
- Should be Preferred by Customer or house owner

Catalogue Objective

To pave way for use of alternate materials and technologies in the reconstruction process as per the principles set by Post Disaster Needs Assessment (PDNA) for housing and human settlements recovery and reconstruction.

Catalogue Limitation

Designs included in this catalogue can be selected and used as they are, for reconstruction of urban and rural housing . For designs, other than those included in this catalogue, detailed engineering design and approval from concerned authorities shall be done.

Lists of Model

1. **Interlocking Brick Masonry One Storey**
   - One Storey
   - One Storey
   - Two Storey
2. **Confined Hollow Concrete Block Masonry**
   - Two Storey
3. **Hollow Concrete Block Masonry**
   - Two Storey
4. **Compressed Stabilized Earth Block Masonry**
   - One Storey
   - Two Storey
5. **Random Rubble Masonry in Mud mortar with GI wire Containment**
   - One Storey
   - Two Storey
6. **Bamboo and Stone Masonry Hybrid Structure**
   - Two Storey
7. **Rat Trap Bond Masonry**
   - One Storey
8. **Earth Bag Masonry**
   - One Storey
9. **Light Gauge Steel Structure**
   - One Storey
   - Two Storey
10. **Steel Structure**
    - Two Storey
11. **Timber Structure**
    - Two Storey
12. **Debris Block Masonry**
    - Two Storey

Model details

1. <u>Interlocking Brick Masonry   One Storey</u>

Interlock Brick Technology consists of specially designed unburnt bricks with tongue and groove features that allows bricks to interlock each other in masonry and thereby reduces mortar usage.

2. <u>Confined Hollow Concrete Block Masonry</u>

Hollow concrete block walls carry the seismic loads and the Reinforced Concrete Columns of minimal size are used to confine the walls.

Model details Contd.

3. Hollow Concrete Block Masonry

Load bearing structure of hollow concrete blocks are seen as a good alternative to conventional brick masonry as they can be locally manufactured, cheaper and environment friendly.

4. Compressed Stabilized Earth Block Masonry

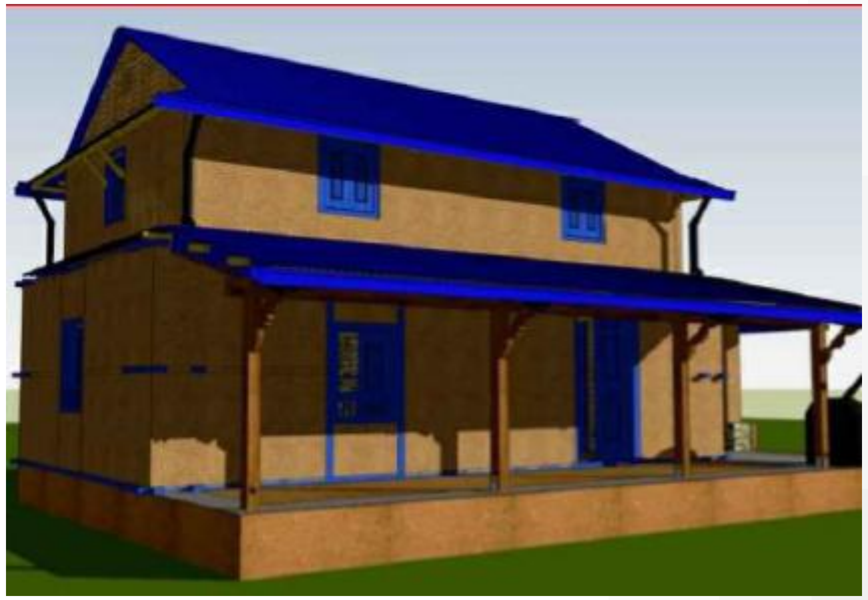
Compressed Stabilized Earth Block **(CSEB)** Technology makes use of mud as a predominant building material. The properties of soil used are improved by using stabilizers like cement. This Block is prepared by machine.

Model details Contd.

5. <u>Random Rubble Masonry in Mud mortar with GI wire Containment</u>

This technology is an improvement on random rubble masonry structure by introduction of GI containment wires. GI wire prevent from flexural failure and delamination of walls.

6. <u>Bamboo and Stone Masonry Hybrid Structure</u>



Local bamboo *(Banbusa Nutans), seasoned and treated, is used in a structural* frame with bamboo wattle and daub panels as walls on the upper floor. The frame is surrounded with a wall in Stone Masonry with Mud Mortar on the ground floor of the house.

Model details Contd.

7. Rat Trap Bond Masonry

Rat-Trap Bond is a modular type of masonry construction in which bricks are laid on edge, thereby creating internal cavity within the wall. The cavity improves the thermal behavior of the wall and significantly reduces the quantity of brick and mortar in the masonry

8. Earth Bag Masonry

Earth bag technology is a simple, inexpensive and sustainable method for building structures using ordinary soil found at construction site.The technology consists of Polypropylene bags filled with locally available soil, laid similarly to masonry with barbed wire serving as a mortar and provides tensile as well as shear strength.

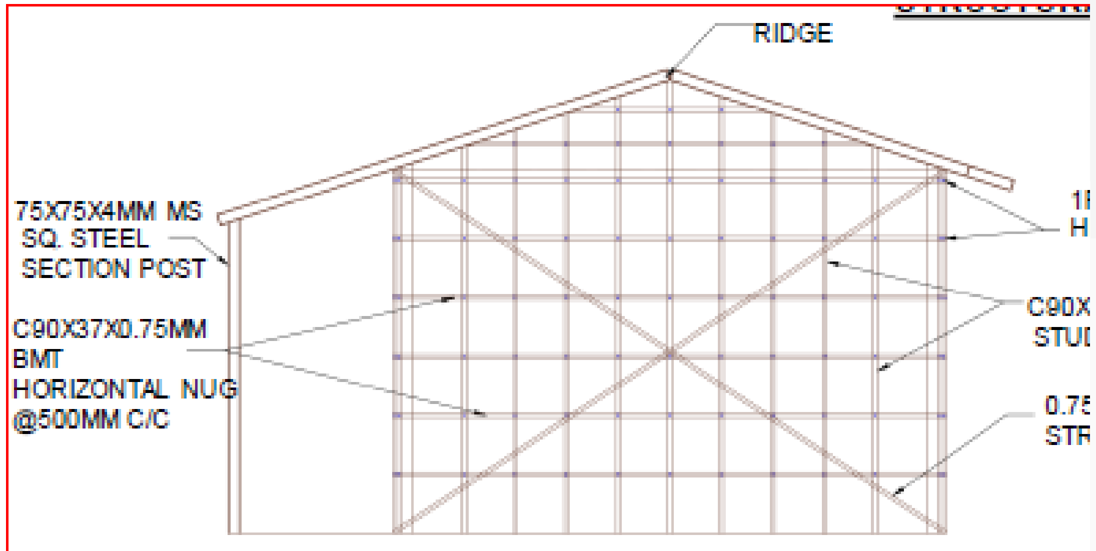
Model details Contd.

9. <u>Light Gauge Steel Structure</u>

It consisting thin steel sections cladded with light gauge steel panel, Cellular light weight concrete, Cement fiber board, gypsum board or calcium silicate board.

10. <u>Steel Structure</u>

It's structural system consisting of mild steel columns and beams to make steel moment resisting frame system. Both the gravity and lateral load is resisted by moment resisting frame. The floor system is made of profile metal decking system over which the thin layer of RCC is laid. The roofing system consists of MS Steel tubes trusswith CGI Sheet.



STRUCTURAL ELEVATION 2

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Model details Contd.

11. Timber Structure

It's structural system consisting of timber studs (vertical members) and horizontal member load bearing system. The timber planks are used as light weight partition walls.

12. Debris Block Masonry

The technology proposes residence construction with block made from stone or brick debris stabilized with cement.

Thank You